Basic Calculus: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

Solution index

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Solution 1 — A simple derivative. Return →

The derivative is $f'(x) = 12x^3 - 2$. At $x = 1$, $f'(1) = 10$. **Check:** the constant term disappears because it does not change with x.

Solution 2 — A definite integral. Return →

An antiderivative is $x^3 + x$. Evaluating from 0 to 2 gives $8 + 2 = 10$. **Check:** the integrand is positive, so the accumulated area is positive.

Solution 3 — Chain rule. Return →

$d(r^3)/dt = 3r^2\, dr/dt$. Here $dr/dt = 4t$, so at $t = 1$, $r = 3$ and the derivative is $3(9)(4) = 108$. **Check:** the outer derivative and inner rate both contribute.

Solution 4 — Small angle. Return →

Convert to radians: $2^\circ = 0.03491$ rad. The approximation gives 0.03491. The exact value is 0.03490, so the agreement is excellent to three significant figures. **Check:** the next term is $-x^3/6 \approx -7 \times 10^{-6}$.

Solution 5 — Exponential decay. Return →

The half-life is $t_{1/2} = \ln 2/\lambda = 6.93$ yr. After 10 yr, the fraction is $e^{-1} = 0.368$. **Check:** 10 yr is about 1.44 half-lives, so a remaining fraction between 1/2 and 1/4 is reasonable.

Solution 6 — Average value. Return →

The average is $(1/3)\int_0^3 x^2 dx = (1/3)(27/3) = 3$. **Check:** the average of x^2 is larger than the square of the average, $(1.5)^2 = 2.25$, because the function is convex.

Solution 7 — A maximum. Return →

Set $dh/dt = 20 - 10t = 0$, giving $t = 2$ s. The height is $h(2) = 40 - 20 = 20$ m. The second derivative is $-10 < 0$, confirming a maximum. **Check:** the projectile returns to the ground at $t = 4$ s, symmetrically.

Solution 8 — Flux from luminosity. Return →

The total energy is $\int_0^{\tau} L_0(1 + t/\tau)dt = L_0[t + t^2/(2\tau)]_0^{\tau} = 1.5 L_0 \tau$. **Check:** the average luminosity over the interval is $1.5L_0$, between L_0 and $2L_0$.

Solution 9 — Taylor approximation. Return →

With $x = 0.04$, $\sqrt{1.04} \approx 1 + 0.02 = 1.02$. The next term is $-x^2/8 = -0.0002$, so the estimate is high by about 2×10^{-4}. **Check:** the exact value is 1.01980, consistent with the estimate.

Solution 10 — A differential equation. Return →

Separate variables: $dy/y = k\,dt$, so $y = y_0 e^{kt}$. Doubling in 5 yr gives $e^{5k} = 2$, hence $k = \ln 2/5 = 0.1386$ yr^{-1}. After 10 yr, $y/y_0 = e^{10k} = 4$. **Check:** two doubling times produce a factor of four.